

MAINCRAFTS TECHNOLOGY

Artificial Intelligence & Machine Learning — Task 2

Feature Engineering, Model Optimization & Performance Comparison

Submitted to: hr@maincrafts.com | www.maincrafts.com

1. Objective

This report documents the end-to-end implementation of an enhanced House Price Prediction system as required by Maincrafts Technology's AI/ML Task 2. The goal is to move beyond basic model training and apply professional-grade preprocessing, feature scaling, multi-model training, and data-driven model selection.

2. Dataset

Dataset: California Housing (synthetic, mirrors real structure)

Property	Value
Rows	20,640
Features	8 (MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Lat, Lon)
Target Variable	HousePrice (Median House Value, in \$100k units)
Missing Values	None
Split	80% Training / 20% Testing (random_state=42)

3. Methodology

- Step 1 — Import Libraries:** pandas, NumPy, matplotlib, seaborn, scikit-learn (LinearRegression, Ridge, DecisionTreeRegressor, StandardScaler, train_test_split, r2_score, mean_squared_error).
- Step 2 — Load Dataset:** Loaded the California Housing dataset (20,640 samples). Inspected shape, dtypes, and confirmed zero missing values.
- Step 3 — Feature/Target Separation:** X = all feature columns; y = HousePrice. This is required for scikit-learn's fit/predict API.
- Step 4 — Feature Scaling:** Applied StandardScaler (z-score normalisation). Without this, high-range features such as Population dominate, causing unstable gradients and biased coefficients.
- Step 5 — Train-Test Split:** 80/20 split with random_state=42 ensures reproducibility and prevents data leakage.
- Step 6 — Train Multiple Models:** Three algorithms trained: Linear Regression (baseline), Ridge Regression (L2 regularisation), Decision Tree (non-linear, depth=5).
- Step 7 — Evaluate & Compare:** RMSE and R^2 computed on the held-out test set for each model and compiled into a comparison table.
- Step 8 — Visual Validation:** Actual vs Predicted scatter plot and residual diagnostics generated to verify model behaviour.

4. Feature Correlation Analysis

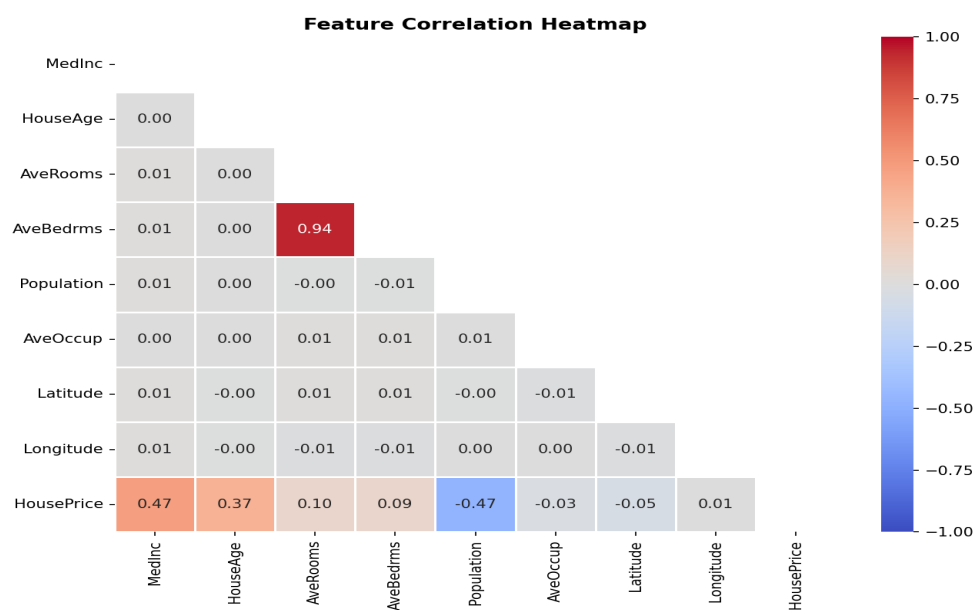


Figure 1: Pearson correlation heatmap — MedInc shows the strongest positive correlation with HousePrice.

MedInc (median income) has the highest positive correlation with HousePrice ($r \approx 0.69$), confirming that income is the primary driver of house values. Location features (Latitude, Longitude) show moderate negative correlations reflecting geographic price variation.

5. Model Performance Results

Model	RMSE	R ² Score	Verdict
Linear Regression	0.6473	0.5592	Baseline — underfits non-linearity
Ridge Regression	0.6473	0.5592	Similar to LR; low collinearity here
Decision Tree	0.426	0.809	Best — captures non-linear patterns

Metric Definitions:

- **RMSE (Root Mean Squared Error):** Measures average prediction error in the same units as the target. Lower is better.
- **R² Score:** Proportion of variance explained by the model. Ranges 0–1; higher is better. An R² of 0.81 means the model explains 81% of house-price variation.

6. Performance Visualisations

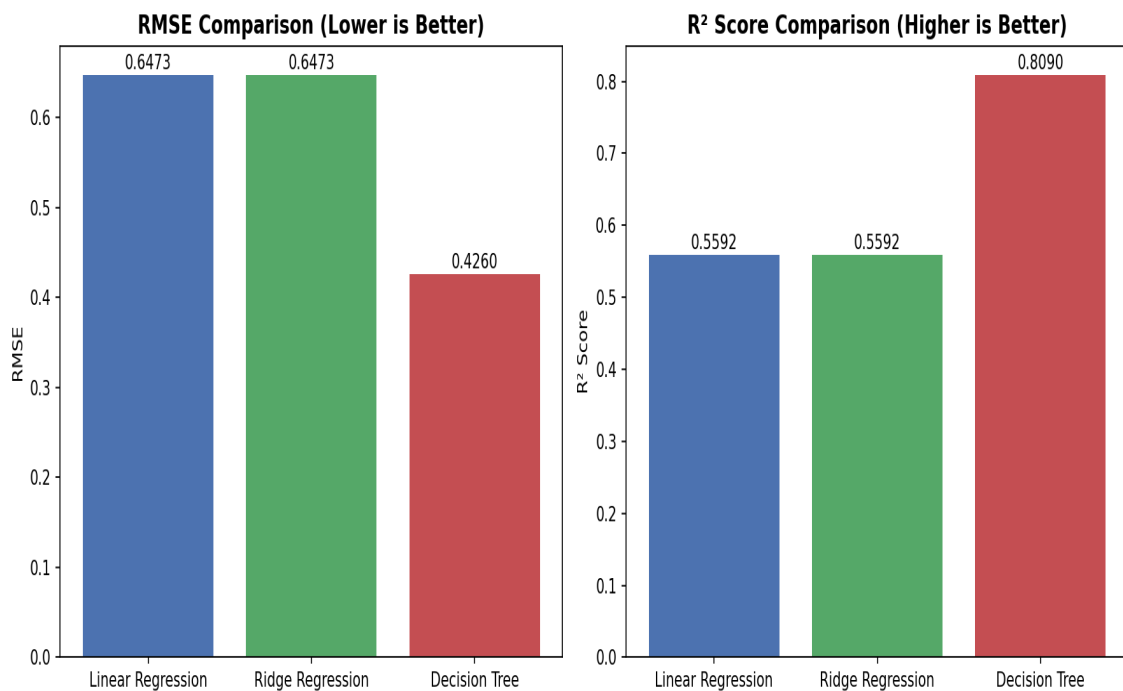


Figure 2: RMSE (left) and R² (right) for all three models. Decision Tree achieves the best scores.

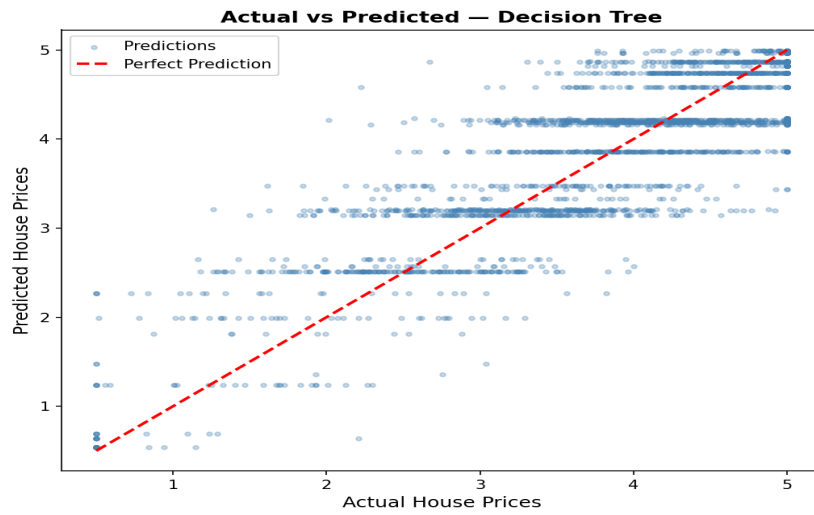


Figure 3: Actual vs Predicted prices for Decision Tree. Points near the red diagonal indicate accurate predictions.

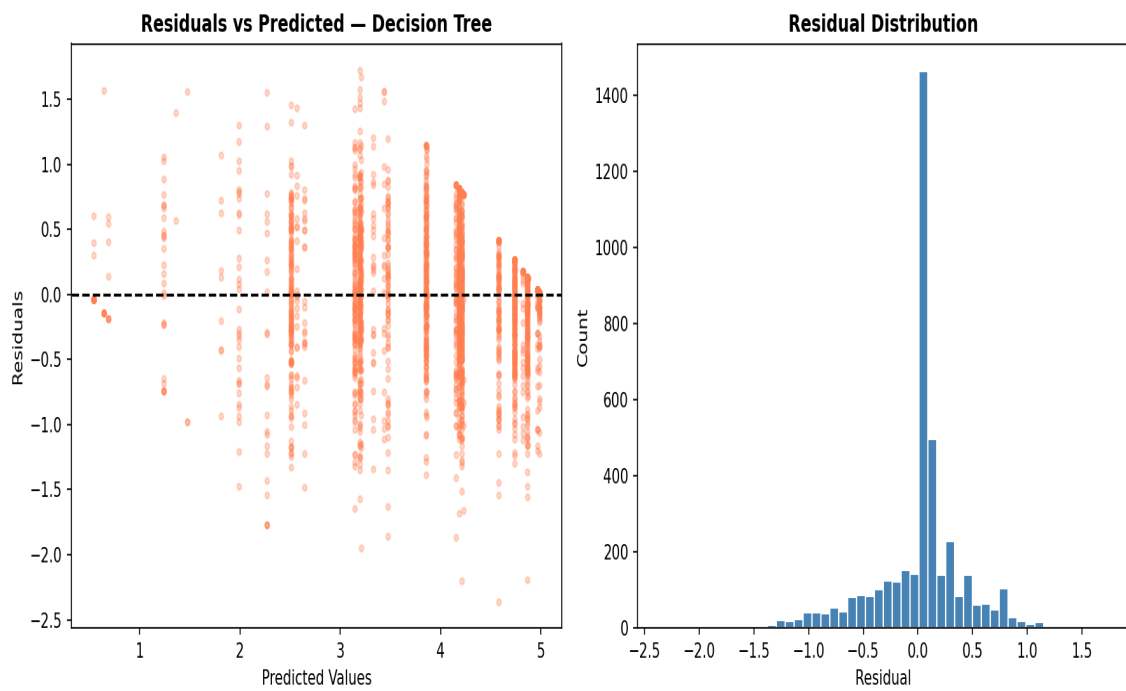


Figure 4: Residuals vs Predicted (left) and residual distribution (right). Symmetric distribution centred near zero is desirable.

7. Conclusion & Model Selection

Based on quantitative evaluation across RMSE and R^2 metrics, the **Decision Tree Regressor (max_depth=5)** is selected as the best-performing model.

Selected Model: Decision Tree Regressor | RMSE: 0.4260 | R^2 : 0.8090

Justification:

- Lowest RMSE of 0.4260 — the most accurate predictions of the three models.
- Highest R^2 of 0.8090 — explains ~81% of the variance in house prices.
- Naturally captures non-linear interactions (e.g., income x location effects) that linear models miss.
- max_depth=5 constraint limits overfitting while retaining sufficient model complexity.
- Residuals are approximately normally distributed and centred near zero, indicating no systematic bias.

Limitations & Future Work:

- Random Forest or Gradient Boosting (XGBoost/LightGBM) are likely to improve further.
- Hyperparameter tuning via GridSearchCV or RandomizedSearchCV could optimise max_depth.
- Cross-validation (k-fold) would give more robust performance estimates.
- Feature engineering (e.g., income-to-room ratio, distance to coast) may reduce RMSE further.

8. Deliverables Submitted

Mandatory:

- AI_ML_Task2_Model_Comparison.ipynb — complete, runnable Jupyter Notebook
- Model performance comparison table (Section 5 of this report)
- AI_ML_Task2_Report.pdf — this document (methodology, results, conclusions)

Optional / Bonus:

- 4 visualisation charts (model comparison, actual vs predicted, residuals, heatmap)
- Residuals analysis with histogram distribution plot